Size effects on bifurcations in nonlinear slots

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Abstract: We study the nonlinear waves in symmetric metal slot waveguides with a Kerr-type core. Using semi-analytical models, we have found new higher order nonlinear modes and studied the influence of slot width on the observed bifurcations.

OCIS codes: 190.6135, 250.5403, 250.4390, 230.7390, 190.3270, 260.5950, 240.6680.

1. Introduction

The studies of nonlinear waves combining both plasmon and soliton features started in 1980s [1, 2]. The interest in the field started to grow again thanks to the results provided in Ref. [3]. Even if no experimental demonstration of plasmon–solitons has been published yet [4], in the mid term, such states can have several applications, e.g. in nonlinear couplers. The structure studied in Ref. [3] and in the present study is a focusing nonlinear dielectric core surrounded by two semi-infinite metal regions (see Fig. 1). It will be called here the nonlinear slot waveguide (NSW). Recently, some stationary waves in such waveguides with a Kerr-type focusing nonlinear core were studied in details [5, 6]. The present work deals with the same type of structures thanks to new models. It has been shown that symmetric, antisymmetric and asymmetric plasmon–solitons can propagate in NSWs. The asymmetric nonlinear mode appears from the symmetric one through a symmetry-breaking bifurcation at a critical power. This bifurcation phenomenon has already been described in several nonlinear waveguides as early as in the eighties (see references in [7]).

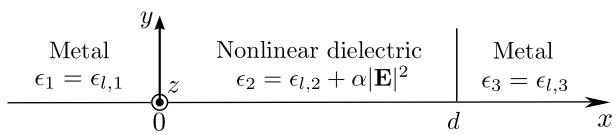

Fig. 1. Nonlinear slot waveguide geometry. The Kerr-type focusing nonlinearity is located only in the dielectric core of width d. The two metal regions are semi-infinite.

2. Methods

The simplified semi-analytical model we developed employs the fact that Maxwell's equations for transverse magnetic electromagnetic waves in the Kerr-type nonlinear medium can be reduced to a single nonlinear wave equation for the magnetic field component [1, 8], assuming that the nonlinear permittivity modification is small and depends only on the transverse electric field component E_x. In a finite size nonlinear core, the field shapes are described by the Jacobi elliptic special functions. This fact combined with the continuity of the tangential field components at the interfaces between nonlinear medium and the bounding metals allows us to obtain the analytical formulæ for the nonlinear dispersion relation.

The second model we proposed extends to the two-interface case the approach developed in [9, 10] for a single nonlinear dielectric/metal interface where the nonlinear term accounts for both transverse and longitudinal electric field components. This model allows to get the dispersion relations separately on the two interfaces of the NSW in an analytical form and to link them by numerical integration of a set of two coupled differential equations in order to calculate the complete NSW dispersion curves and the field shapes.

3. Results

In our study, we obtain a number of previously unknown higher order symmetric, antisymmetric and asymmetric modes in the NSW and compute their dispersion curves (only partial results shown in the inset in Fig. 2 Left)).

Qualitatively the results of the simplified but semi-analytical approach and the exact model are the same. Both our models predict the bifurcation of asymmetric modes from the symmetric node-less modes (see Fig. 2 Left).

We show that the bifurcation occurs not only for the first symmetric mode (coming from the fundamental linear mode, blue curve with flat part) as it is already known [5] but also for higher order nonlinear modes (see inset in Fig. 2 Left). Finally, using the ability of our models to tackle finite size nonlinear region, we show that the first nonlinear asymmetric mode dispersion curves have invariant parts with respect to the core width (see Fig. 2 Right). We describe accurately these parts using analytical formulæ based on a simple physical model.

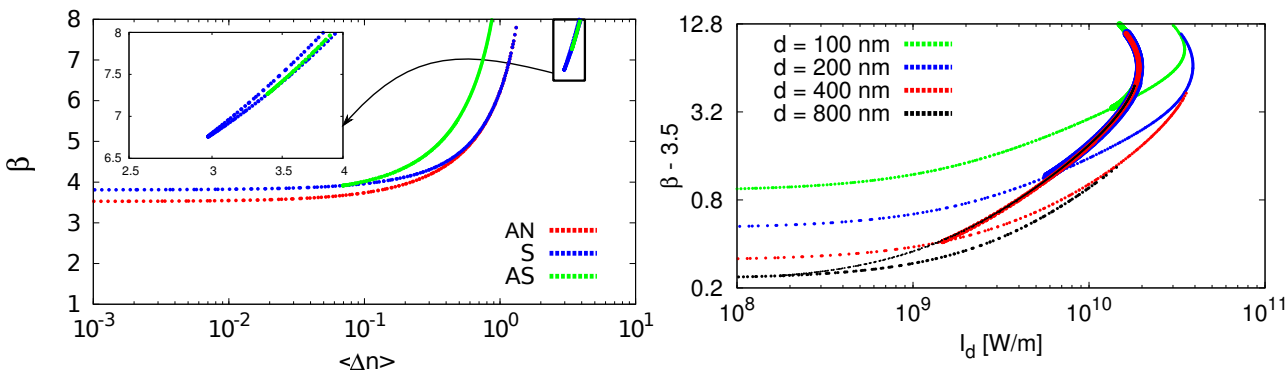

Fig. 2. Left: Dispersion curves of the first and higher order symmetric (S — blue), antisymmetric (AN — red), and asymmetric (AS — green) modes for the symmetric NSW obtained with our 2nd model as a function of the spatial average of the refractive index induced by the nonlinearity <Δn>. The inset shows a zoom on the region with the bifurcation of the first symmetric higher order mode. Results obtained for a 400-nm-wide hydrogenated amorphous silicon core $\varepsilon_{l,2} = 3.46^2$ ($n_2 = 2 \cdot 10^{-17}$ m^2/W) sandwiched between gold $\varepsilon_1 = \varepsilon_3 = -90$ at a $\lambda = 1.55$ μm.
Right: Locus of the asymmetric mode dispersion curves for various core widths d as a function of light intensity in the core. Only first symmetric (thin curve) and asymmetric modes (thick curves) are shown using one color per width d. Both axes are in log scale.

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